

Research Article

Powder plasma arc additive manufacturing of CoCrFeNiW_x high-entropy alloys: Microstructure evolution and mechanical propertiesQingkai Shen, Jiaxiang Xue^{*}, Xiaoyan Yu, Zehong Zheng, Ning Ou

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ABSTRACT

Powder plasma arc additive manufacturing (PPA-AM) is a promising method for the preparation of high-entropy alloys (HEAs), and has the advantages of low cost, high efficiency, and high quality. In this study, the CoCrFeNiW_x ($x = 0, 0.2, 0.5, 0.7$, and 1.0) HEAs were successfully prepared via the PPA-AM technique. Microstructure and mechanical properties of alloys with different W content were investigated. The results showed that the microstructure changed from a single FCC phase ($x = 0$) to the dual-phase FCC + μ ($x = 0.2$ and 0.5), and then to a three-phase combination of FCC + μ + BCC ($x = 0.7$ and 1.0) with the addition of W. The volume fraction of the μ phase and BCC phase increases with the increase in W content, while the size of FCC columnar grain is significantly reduced. The solid-solution strengthening of the FCC phase and the second phase strengthening were mainly responsible for the increase in the microhardness (136–413 HV) and yield strength (187–567 MPa) while the brittleness of the μ phase contributes to the expansion of the crack, which leads to a decrease in the ductility (50.8% to 0.3%). These results can provide a systematic understanding of the phase transformation of PPA-AM HEAs with different W content and its effects on the mechanical properties.

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1. Introduction

High-entropy alloys (HEAs), also called multicomponent alloys, were proposed by Yeh and Cantor [1,2] in 2004. Different from traditional alloys, HEAs are usually composed of four or more principal elements, and the content of each element is between 5 at% and 35 at% [3]. Due to the unique composition of these HEAs, the mixing entropy of the alloys is high [1], which will inhibit the formation of intermetallic [4] and prefer to form stable and simple solid solution phases [5]. In addition to the high entropy effect, HEAs are also subject to the lattice distortion effect [6], slow diffusion effect [7], and cocktail effect [8]. These four effects result in HEAs with excellent properties such as outstanding mechanical properties [9,10], high temperature resistance [11–13], corrosion resistance [14,15] and radiation resistance [16,17], when their composition is designed appropriately. Therefore, HEAs have great application potential in aerospace, transportation, petrochemical, and nuclear industries.

As the demand for metallic materials of different property profiles increase, the development of HEAs with excellent properties has always been a popular research direction. The strengthening

mechanisms of the HEAs mainly include fine grain strengthening, solid solution strengthening, second phase strengthening, and deformation strengthening. Chen et al. [18] proved that the grain size of HEAs can be significantly reduced by rapid cooling by facilitating fine grain strengthening. Solid solution strengthening is a common strengthening method used in HEAs [19]. The dissolution of additive elements with large atomic radii in HEAs cause solid solution strengthening and increase the strength of the alloys [20]. The typical method is to add large-radius Al or Ti to CoCrFeNi [21,22]. The Co, Cr, Fe, and Ni have similar atomic radii and can form stable FCC solid solution phase [23]. Adding large-radius atoms causes lattice distortion, which will produce a solid solution strengthening effect and improve the strength of the alloy [24]. In addition, the second phase strengthening also plays a very important role in the strengthening of HEAs. For example, in CoCrFeNi, when the Al content reaches a certain percentage, the alloy changes from the FCC phase to the FCC + BCC dual phase, and then forms a single-phase BCC alloy [25]. Due to the appearance of the BCC phase, the strength of the alloy is improved significantly [26]. Deformation strengthening occurs mainly through cold rolling, hot rolling, or forging to deform the alloys, resulting in work hardening and fine grain structure to strengthen the alloy significantly [27].

HEAs are usually prepared by arc melting and induction melting followed by casting [28]. However, as HEAs are composed of a

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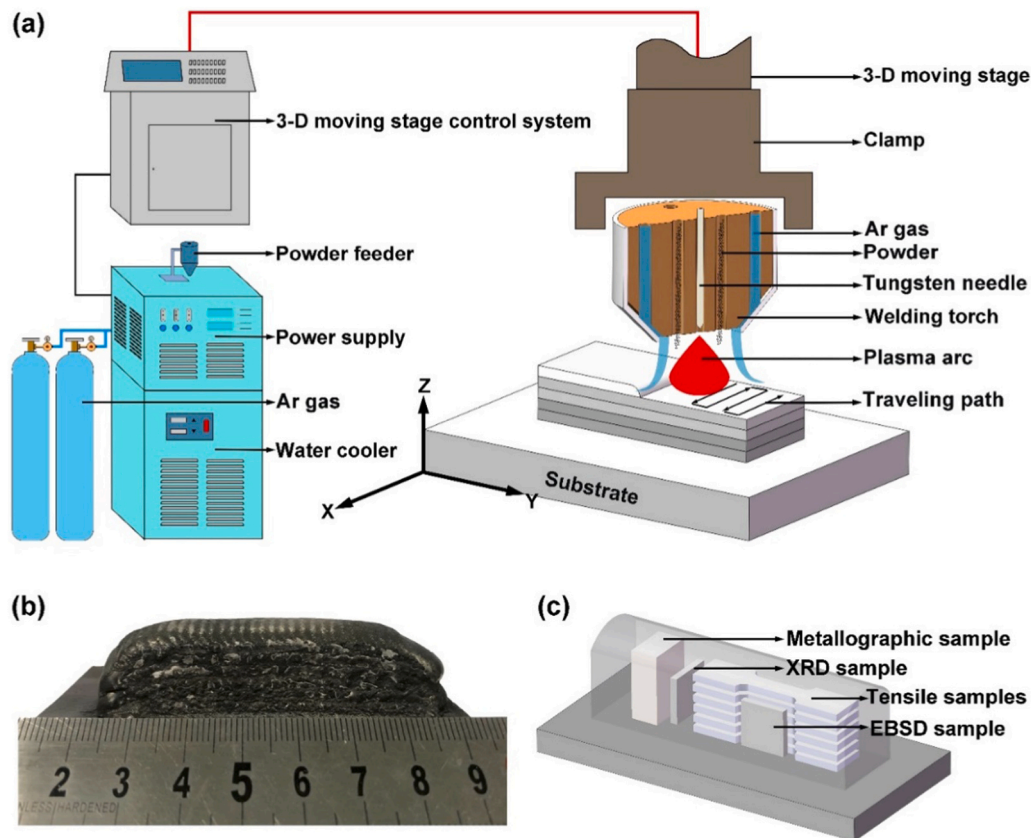


Fig. 1. (a) Schematic diagrams of the PPA-AM process setup, (b) photo of the $W_{1.0}$ sample, (c) schematic of samples extraction for tests.

variety of metal elements, homogenous dissolution is not easy, and component segregation often occurs [29]. Additive manufacturing (AM) technology is used to stack melted materials layer by layer into required shapes via computer control. AM technology has become a popular part fabrication technology that has the advantages of flexibility, high efficiency, and the ability to realize complex shape parts. This manufacturing method has been widely used in the field of HEAs. The common AM methods for HEAs are selective laser melting (SLM), selective electron beam melting (SEBM), and laser melting deposition (LMD). Compared with conventional alloys prepared by melting, alloys prepared by AM methods usually have finer grains and better comprehensive mechanical properties [30]. This is due to the large temperature gradients and rapid solidification during AM process [31], which reduces the grain size and retains the homogeneity of the alloy [32]. In recent years, plasma arcs have been used as heat source to melt metal powder for the HEA additive manufacturing. Compared with the SLM method, plasma arc has the characteristics of high heat input and a pulse-stirred molten pool, and especially for alloys with high melting points, the temperature of the plasma arc is higher, which can fuse the mixed powder more fully. Therefore, plasma arc additive manufacturing has become a potential AM method for HEAs [33,34].

In this work, powder plasma arc additive manufacturing (PPA-AM) is employed for the AM of $CoCrFeNiW_x$ ($x = 0, 0.2, 0.5, 0.7$ and 1.0) HEAs with different W content. The effect of W content on the microstructure and mechanical properties were discussed systematically. In addition, the strengthening mechanisms and ductility reduction were also investigated.

2. Experimental procedures

In this work, pure Co, Cr, Fe, Ni, and W powders (99.5% purity) were prepared by high-pressure gas atomization with a diameter of

48–75 μm . Compared with the pre-alloyed powder, the use of the mixed elemental powder can not only reduce the powder cost, but also facilitate a flexible alloy composition. The powders were used to deposit $CoCrFeNiW_x$ ($x = 0, 0.2, 0.5, 0.7$, and 1.0 , x values in molar ratio) HEAs to form thick walls. To obtain a homogeneous powder blend, a high-energy ball mill (YQM-2 L) was employed for 8 h. The Al_2O_3 ball with a diameter of 10 mm was used as the process medium, and the weight ratio of the ball to powder is 2:1. The PPA-AM processing of $CoCrFeNiW_x$ alloys was conducted on a PPA-AM platform (DML-V03AD, Duomu) as shown in Fig. 1(a). The PPA-AM process was conducted under an oscillation scanning path, with a plasma arc current of 80 A and scan rate of 5 mm/s using high purity Ar shielding gas of 15 L/min. The detailed processing parameters are listed in Table 1. Six-layer thick walls were deposited to the approximate sizes of 60 mm (length) $\times$ 15 mm (breadth) $\times$ 18 mm (height) on a 304 substrate (as shown in Fig. 1(b)). The $CoCrFeNiW_x$ ($x = 0, 0.2, 0.5, 0.7$ and 1.0) samples were denoted as W_0 , $W_{0.2}$, $W_{0.5}$, $W_{0.7}$, and $W_{1.0}$, respectively, and the nominal composition of the alloys are shown in Table 2.

Fig. 1(c) shows the schematic of sample positioning for microstructural and mechanical tests. The phase structures of the $CoCrFeNiW_x$ HEA samples were determined by a D/Max-2600 X-ray

Table 1
Processing parameters.

Parameters	Value	Unit
Scan rate	5	mm/s
Current range	80	A
Shielding gas flow	15	L/min
Layer thickness	~3	mm
Wait time between layer	3	min
Contact tip to work distance	8–10	mm

Table 2
Chemical composition of CoCrFeNiW_x samples (at%).

Alloy	Positions	W	Co	Cr	Fe	Ni
W ₀	Nominal	0	25.00	25.00	25.00	25.00
	FCC	0	25.51	23.26	26.45	24.77
W _{0.2}	Nominal	4.76	23.81	23.81	23.81	23.81
	FCC	3.73	24.75	22.08	25.96	23.49
W _{0.5}	μ	11.44	19.62	30.00	21.01	17.94
	Nominal	11.11	22.22	22.22	22.22	22.22
W _{0.7}	FCC	7.45	22.17	20.28	23.49	24.07
	μ	35.14	18.57	18.23	16.89	11.17
W _{1.0}	Nominal	14.89	21.28	21.28	21.28	21.28
	FCC	7.47	22.05	19.99	24.09	26.40
	μ	37.16	18.31	16.42	16.62	11.47
	BCC	100	–	–	–	–
	Nominal	20.00	20.00	20.00	20.00	20.00
	FCC	7.78	21.98	18.23	24.79	27.23
	μ	37.50	18.21	15.92	17.01	11.36
	BCC	100	–	–	–	–

diffractometer (XRD) with Cu K α radiation at 40 kV and 150 mA, at a scanning speed of 6°/min. The XRD test samples were polished to #2000 with SiC paper. An FEI Nova NanoSEM430 scanning electron microscope (SEM) equipped for energy-dispersive X-ray spectroscopy (EDS) and electron backscatter diffraction (EBSD) detection was used to characterize the microstructures of the alloys. The metallographic samples were ground with SiC paper and polished with a 0.25 μ m diamond suspension for 10 min. The Vickers micro-hardness measurement was carried out on a Shimadzu HVM-2 T micro-hardness tester. The hardness tests were performed at 1 mm intervals along the centerline from the top to the bottom of the samples, under a load of 9.8 N with a dwell time of 15 s. The tensile tests of the CoCrFeNiW_x HEAs were performed on an Instron 3369 testing machine at room temperature with a strain rate of 10^{−3} s^{−1}.

3. Results and discussion

3.1. Microstructures

The XRD patterns of the PPA-AM CoCrFeNiW_x ($x = 0, 0.2, 0.5, 0.7$, and 1.0) HEAs are shown in Fig. 2(a). According to the XRD patterns, a single FCC phase appeared in the W₀ alloy. With the addition of W, minor diffraction peaks corresponding to the μ (Fe₇W₆-type) phase in the W_{0.2} alloy were detected, which was consistent with previous reports [35,36]. More diffraction peaks of the μ phase were clearly observed in W_{0.5}, W_{0.7}, and W_{1.0} alloy, and with the increase in the W content, the peak intensities of the μ phase also increased. It suggests that addition of W can promote the formation of the μ phase. When the W content was increased to 0.7 and 1.0, new peaks of the BCC phase appeared in the W_{0.7} and W_{1.0} alloys.

The enlarged XRD patterns of the (200)_{FCC} peak between 48° and 54° are shown in Fig. 2(b). Obviously, for $x = 0 - 0.7$, the (200)_{FCC} peak shifted gradually towards the lower 2 θ angle with the increase in W content. As the W content in the alloy is greater than 0.7, the (200)_{FCC} peak no longer shifts. The lattice parameters of the FCC phase were calculated according to Bragg's Law. As shown in Fig. 2(c), the lattice parameter of the FCC phase increases initially until $x = 1.0$ (from 3.585 Å (W₀) to 3.605 Å (W_{0.7} and W_{1.0})). This trend can be attributed to the presence of W, with a larger atomic radius (136.7 pm) relative to the Co (125.1 pm), Cr (124.9 pm), Fe (124.1 pm), and Ni (124.6 pm) [23], dissolved into CoCrFeNi solid solution, resulting in the lattice distortion and increase of the lattice parameters. In addition, when the W content varies from 0.7 to 1.0, the lattice parameters do not change. This is mainly because of the limited solubility of W in the FCC phase, which results in the accumulation of W in the FCC to form a W phase and μ phase.

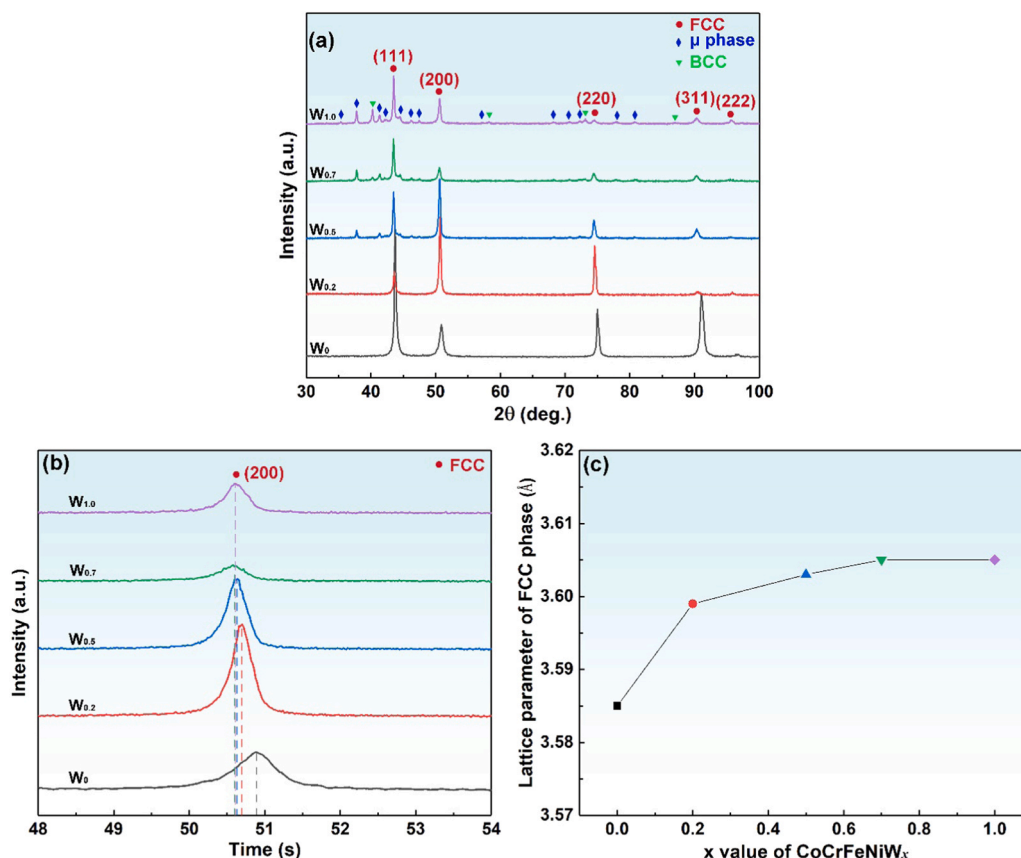


Fig. 2. (a) XRD patterns of CoCrFeNiW_x HEAs, (b) enlarged image of the FCC(200) phase and (c) lattice parameters of FCC phase with increasing W content.

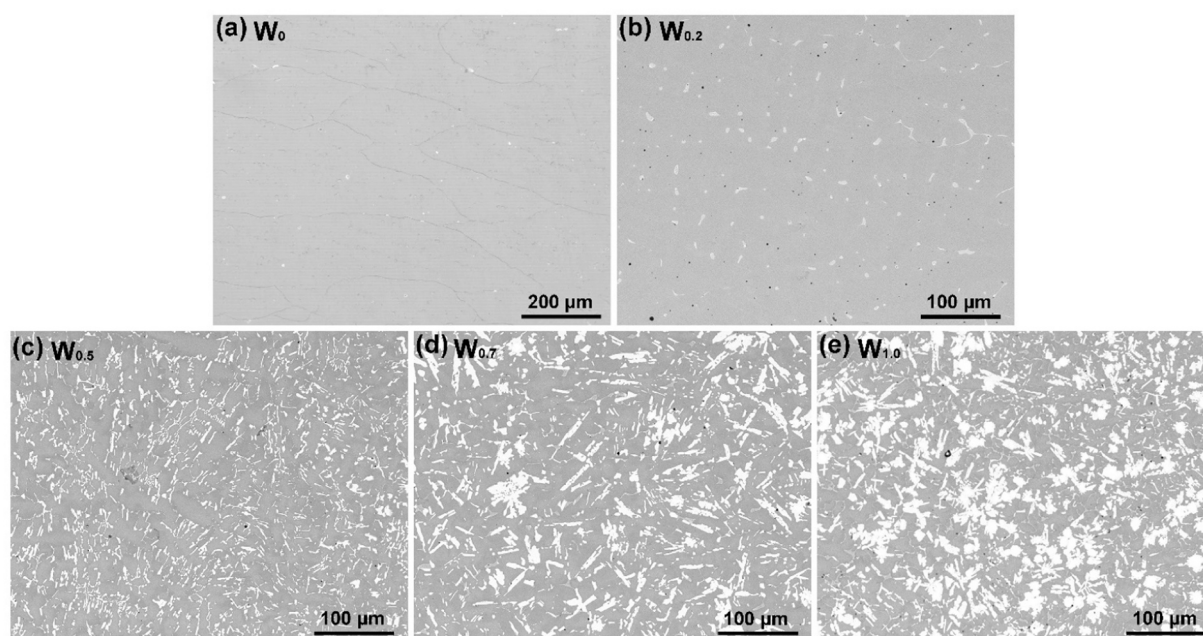


Fig. 3. SEM images of PPA-AM CoCrFeNiW_x HEAs (a) W₀, (b) W_{0.2}, (c) W_{0.5}, (d) W_{0.7} and (e) W_{1.0}.

The SEM images of PPA-AM CoCrFeNiW_x HEAs are shown in Fig. 3 and Fig. 4. The microstructure of the W₀ alloy exhibiting a single FCC phase has a typical columnar grain morphology, and the average grain is approximately 200 μm in width and 2 mm in length (Fig. 3(a)). The formation of columnar crystals may be attributed to the high solidification rates and sharp thermal gradients of PPA-AM, and this phenomenon is similar to the laser additive manufacturing [37]. By adding W, some white precipitates with a diameter of 5–10 μm (identified as μ phase) were observed in the W_{0.2} HEA (Fig. 3(b)). Fig. 4(a) shows the magnified images of the W_{0.2} alloy and revealed that the μ phase presented a network structure. With the

increase in W content, the volume fraction of white precipitates in the alloy increases accordingly (as shown in Fig. 3(c-e)). In alloy W_{0.5}, the μ phase exhibited a lamellar morphology (about 10 μm in length) that was evenly distributed in the FCC matrix, indicating the hypoeutectic structure of W_{0.5}. As shown in Fig. 3(d), the μ phase precipitates occur in the W_{0.7} alloy with a length of about 30–70 μm. In addition, irregular massive precipitation appeared in the alloy (Fig. 4(c)), which corresponds to the BCC phase mentioned in XRD patterns. Under the dilution of the FCC matrix, the μ phase was formed around the BCC phase. With the addition of more W, the precipitates were further coarsened and many massive BCC phases

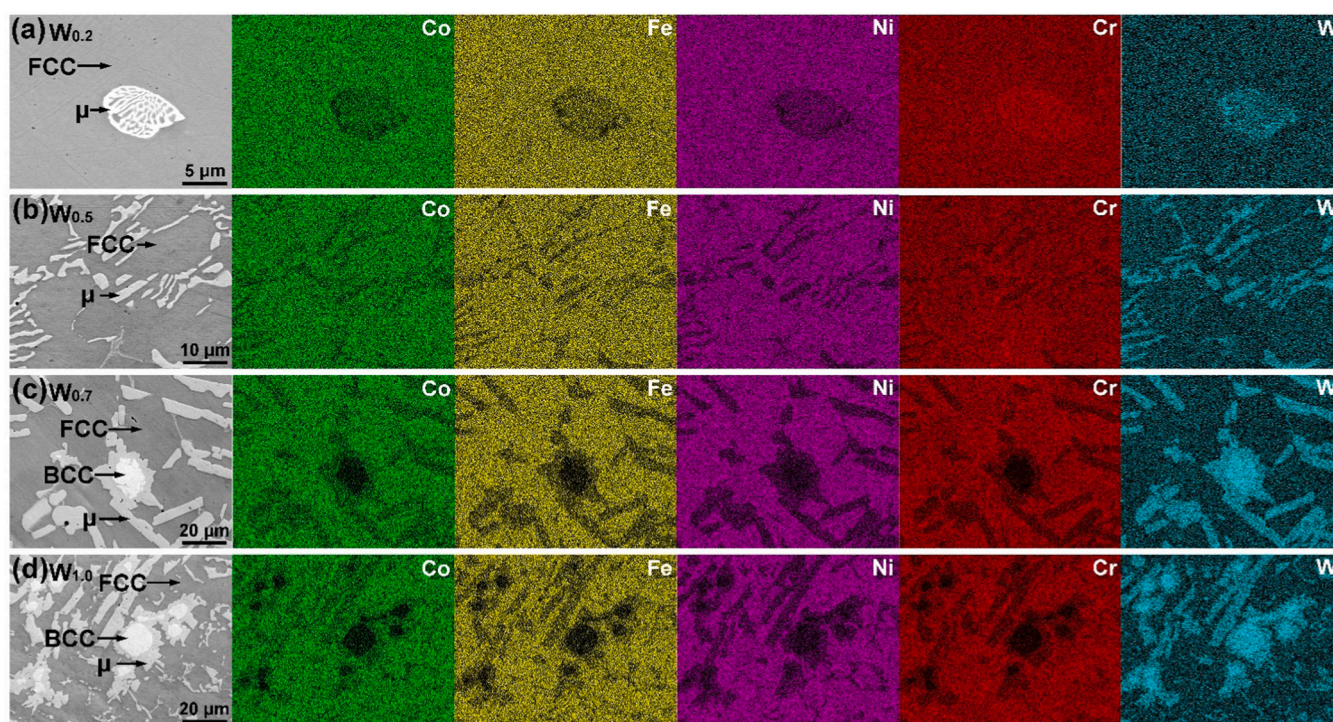


Fig. 4. EDS mappings of PPA-AM CoCrFeNiW_x HEAs (a) W_{0.2}, (b) W_{0.5}, (c) W_{0.7} and (d) W_{1.0}.

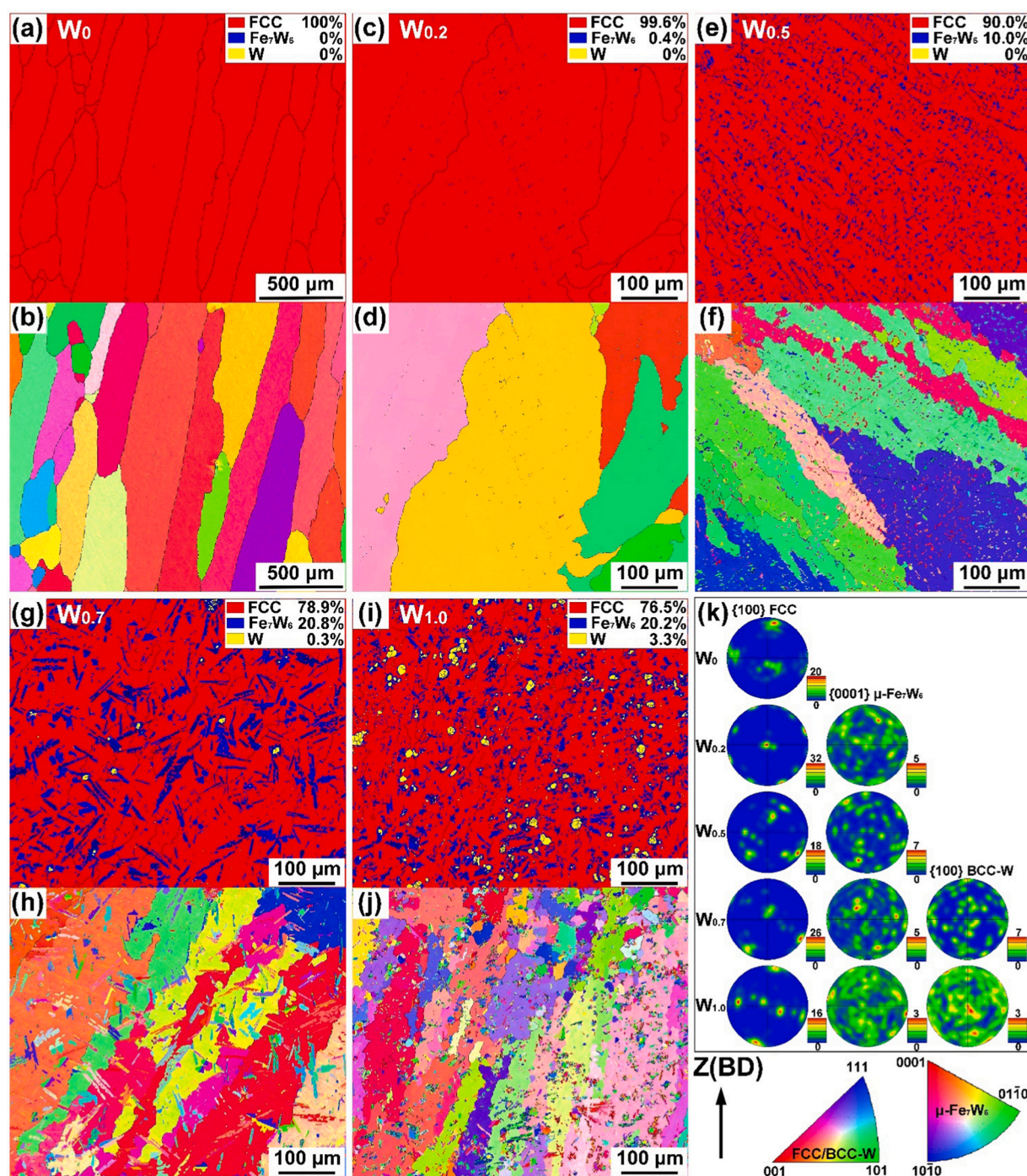


Fig. 5. EBSD data of CoCrFeNiW_x HEAs with different W addition: phase map and IPF (a, b) W₀, (c, d) W_{0.2}, (e, f) W_{0.5}, (g, h) W_{0.7} and (i, j) W_{1.0}; (k) PF of the HEAs.

with a diameter of $\sim 20\text{ }\mu\text{m}$ were formed, as shown in Fig. 3(e) and 4(d), and a fine needle-like μ phase was precipitated in the FCC matrix. To sum up, the microstructure of the CoCrFeNiW_x alloys changed from the single FCC to the duplex FCC + μ phase and finally, to the three-phase FCC + μ + BCC with the addition of W.

The EDS maps of PPA-AM CoCrFeNiW_x HEAs are shown in Fig. 4, and the corresponding chemical composition of the different regions are summarized in Table 2. The chemical composition of the W₀ alloy was close to the nominal composition despite a slight drop in the Cr content for the as-deposited alloy; this drop was a common occurrence in the case of AM alloys [38,39]. The elemental distributions of the μ phase in the W_{0.2} alloy are shown in Fig. 4(a). It revealed that the μ phase is enriched with W and Cr. However, the

chemical composition of the μ phase in $W_{0.5}$, $W_{0.7}$, and $W_{1.0}$ alloys exhibited higher concentrations of W (as shown in Fig. 4(b-d) and Table 2). The solubility of W in CoCrFeNi is limited (about 4 at%), and higher concentration leads to an Fe_7W_6 -type peritectic reaction and the formation of the μ phase. This result agrees with the previous work by Wu et al. [40] and Wang et al. [36]. The Cr enrichment in the $W_{0.2}$ alloy is mainly because the Cr in the CoCrFeNi alloys were not thermodynamically stable and tends to form Cr metal clusters [41]. The μ phase boundaries provide prolonged diffusion pathways for the Cr atoms [42,43], the boundaries of the μ phase are enriched with Cr. This phenomenon occurs in all alloys with a precipitated μ phase. Meanwhile, the EDS results of the $W_{0.7}$ and $W_{1.0}$ alloys (Table 2) indicated that the BCC phase was pure W, which may be

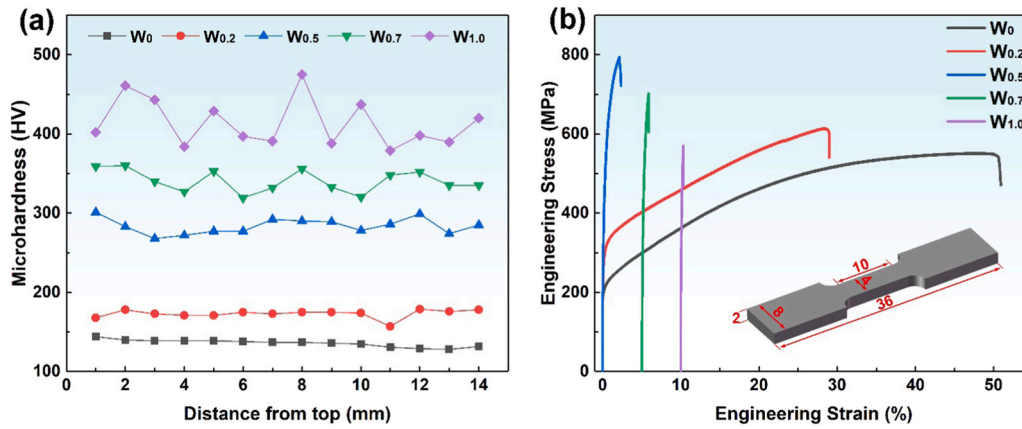


Fig. 6. (a) Microhardness curves along the build direction and (b) tensile strain-stress curves of the PPA-AM CoCrFeNiW_x HEAs.

attributed to the solubility limit of W in the FCC and μ phases. Pegues et al. [44] reported that the pure W-BCC phase appeared in the range of 10–21 at% W content in CoCrFeMnNi, which is almost consistent with this study. Furthermore, the EDS results showed that the W dissolved in the FCC phase, causing an increase in the lattice parameters (Fig. 2(c)). However, the dissolution of W content in the FCC phase was limited at about 7.5 at%, which caused the lattice parameters of the FCC phase in W_{0.5}, W_{0.7}, and W_{1.0} alloys to change little.

Fig. 5 shows the EBSD data on the cross-section of CoCrFeNiW_x HEAs along the build direction (BD). The phase maps and inverse pole figure (IPF) maps are shown in Fig. 5(a–j). Obviously, with the increase in W content, the proportion of precipitates (μ and BCC phase) in the alloy increases gradually. When x increases from 0.7 to 1.0, the proportion of the μ phase does not increase, remaining constant at approximately 20%, while the proportion of the BCC phase in the alloy increases from 0.3% to 3.3%. In addition, the FCC grain size decreased significantly after the addition of W. Due to the lack of grain nucleation during solidification and the high energy input of the PPA-AM process, the W₀ alloy forms extraordinarily large columnar grains [45]. On the one hand, the grain refinement is due to the high melting point of W, which tends to solidify preferentially in the liquid phase [46]. The preferentially precipitated BCC phase and μ phase provide nucleation for the growth of the FCC phase. On the other hand, the pinning effect of the larger μ phase and BCC phase particles hinder the growth of the FCC phase in W_{0.7} and W_{1.0}, which significantly reduces the size of FCC [47].

The IPF maps show that the FCC phase prefers to grow along the $\langle 100 \rangle$ direction. Not surprisingly, the FCC phase is easily affected by heat flows and grows along the direction of maximum heat extraction in the AM process [48]. During the PPA-AM process, the direction of the maximum temperature gradient is perpendicular to the melt pool. However, the growth direction of columnar FCC grains is not completely parallel to the BD. This is because the PPA-AM process was conducted with an oscillating scanning path, which changed the local heat transfer direction, resulting in an inclined grain growth direction. The corresponding pole figure (PF) maps are shown in Fig. 5(k). The PF reveal that the FCC phase has a strong $\{100\}$ texture, while the BCC phase and μ phase are less affected by heat, showing random growth.

3.2. Mechanical properties

The microhardness curves of the PPA-AM CoCrFeNiW_x HEAs are shown in Fig. 6(a) and the average hardness values are listed in Table 3. Obviously, the W content has a significant influence on the hardness of the CoCrFeNiW_x alloys. With the increase in the W content, the average hardness values increased from 136 HV (W₀) to

413 HV (W_{1.0}). In addition, the fluctuation of hardness value also increases gradually with the increase in W content in the alloy (as shown in Fig. 6(a)). The fluctuation of hardness was attributed to the increase in the volume fraction of the μ phase and BCC phase, which means that the hardness test points are on different phases, resulting in a difference in the hardness values.

The typical engineering strain-stress curves of the CoCrFeNiW_x HEAs with different amounts of W are shown in Fig. 6(b). The tensile yield strength (YS), ultimate tensile strength (UTS), and elongation to failure (EF) are summarized in Table 3. The W₀ alloy exhibited an excellent EL of 50.8% with relatively lower YS (186.8 MPa) and UTS (526.6 MPa). With the increase in the W content in CoCrFeNiW_x HEAs, the tensile strength of the alloy increases, while the plasticity decreases. The W_{0.2} HEA has a significantly increased YS of 284.8 MPa while the UTS increased to 627.4 MPa and the EL reduced sharply (28.7%). When the W content exceeded 0.5, the YS and UTS were considerably increased to 461.9 MPa and 786.8 MPa, respectively, with a further reduced EL of 2.6%. As the W content increased further, the W_{0.7} and W_{1.0} alloys were too brittle. The YS and UTS of the W_{0.7} alloy are 555.4 MPa and 597.8 MPa, respectively, but the EL was only 0.6%. In the tensile test, the yield and fracture of the W_{1.0} alloy occur almost simultaneously (566.7 MPa), and the elongation is 0.3%.

The significant increase in the hardness and YS of PPA-AM CoCrFeNiW_x HEAs were attributed to two reasons. Firstly, the dissolution of W atoms in the FCC phase increased the lattice distortion of FCC, which induced solid-solution strengthening. Secondly, the formation of the W-enriched μ phase and BCC phase significantly enhanced the strength of the CoCrFeNiW_x alloys due to the second phase strengthening. As presented in Table 2, the W contents in the FCC phase in W_{0.2}, W_{0.5}, W_{0.7}, and W_{1.0} were 3.73 at%, 7.45 at%, 7.47 at%, and 7.78 at%. The lattice parameters of the FCC solid solution were 3.585 Å, 3.599 Å, 3.603 Å, 3.605 Å, and 3.605 Å in the CoCrFeNiW_x ($x = 0, 0.2, 0.5, 0.7$, and 1.0) alloys, respectively. For W_{0.2}, the volume fractions of the μ phase were relatively low, the improvement in strength was attributed to solid solution strengthening of FCC. In the W_{0.5} alloy, the solid solution strengthening of the FCC

Table 3
Mechanical properties of the PPA-AM CoCrFeNiW_x HEAs.

Alloys	Hardness (HV)	Yield strength YS (MPa)	Ultimate tensile strength UTS (MPa)	Elongation to failure EF (%)
W ₀	136.3 ± 4.4	186.8 ± 16.6	526.6 ± 37.9	50.8 ± 2.3
W _{0.2}	172.7 ± 5.3	284.8 ± 6.7	627.4 ± 18.3	28.7 ± 3.3
W _{0.5}	283.5 ± 9.9	461.9 ± 10.3	786.8 ± 30.7	2.6 ± 1.0
W _{0.7}	338.8 ± 14.3	555.4 ± 14.8	597.8 ± 73.8	0.6 ± 0.2
W _{1.0}	413.4 ± 30.5	566.7 ± 3.2	566.7 ± 3.2	0.3 ± 0.02

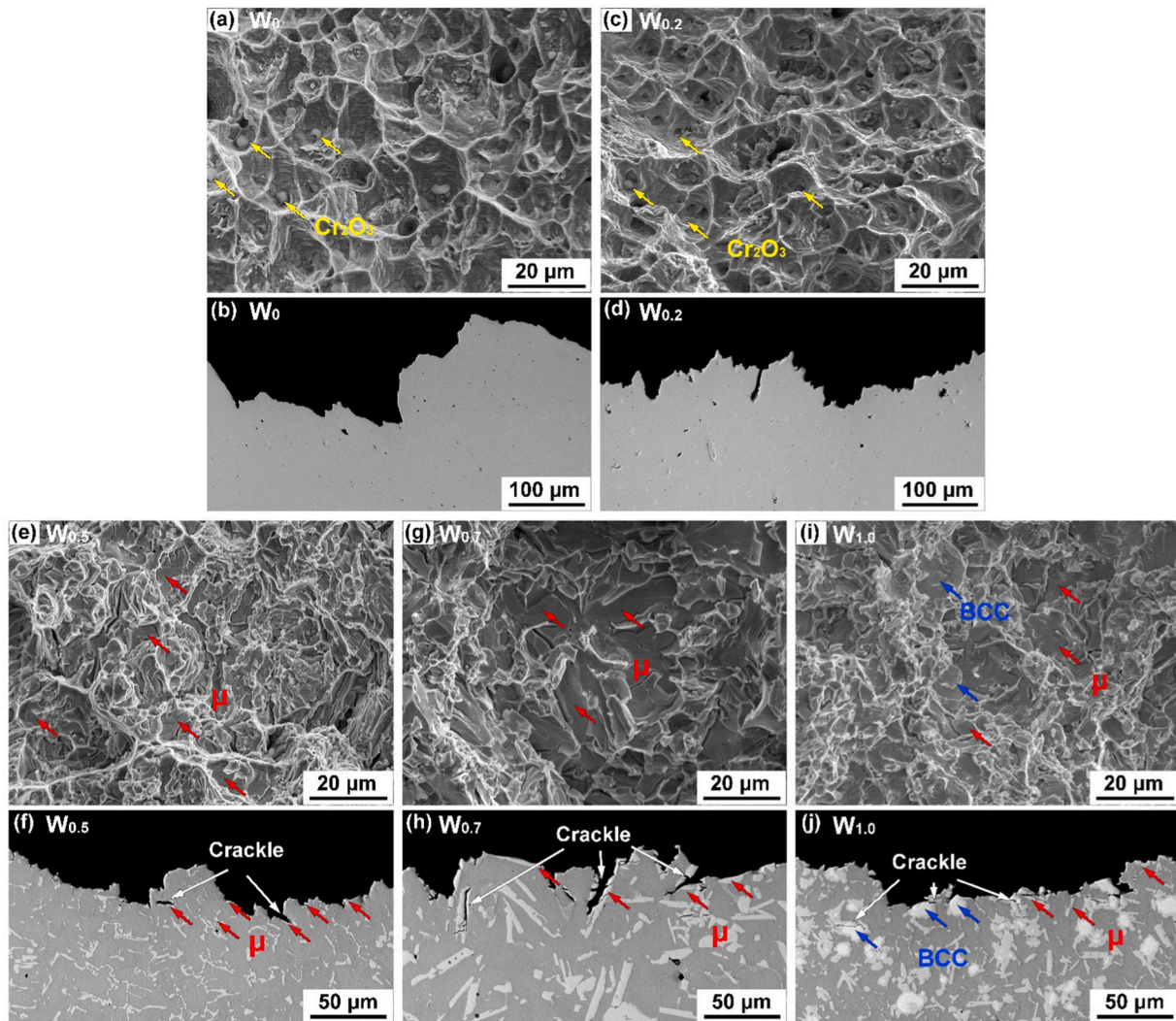


Fig. 7. Representative fractograph of CoCrFeNiW_x HEAs with different W addition (a, b) W₀, (c, d) W_{0.2}, (e, f) W_{0.5}, (g, h) W_{0.7} and (i, j) W_{1.0}.

phase and the precipitation of the μ phase caused the improvement in strength. Besides, the μ phase was a hard intermetallic indexed as the hexagonal structure [49], and led to a decrease in the plasticity of W_{0.5} alloy. The improvement in solid-solution strengthening was limited by the solubility of W atoms in the FCC phase. Therefore, the strengthening of the W_{0.7} and W_{1.0} alloys was mainly due to the precipitation of the μ phase and BCC phase, further embrittling the alloys.

To further understand the influence of W addition on the fracture mechanism of PPA-AM CoCrFeNiW_x HEAs, the tensile fracture surface and lateral surfaces of fracture have been characterized by SEM (Fig. 7). For the W₀ alloy sample, deep dimples were observed on the fractured surface, which indicated that the alloy undergoes ductile fracture (as shown in Fig. 7(a)). Sub-micron particles were observed at the bottom of some dimples, and an EDS analysis revealed that the particle is mainly rich in Cr and O. It was easier to oxidize the Cr in AM processing [50], so the particles should be Cr₂O₃. The fracture surface of the W_{0.2} alloy was like that of the W₀ alloy and showed a typical ductile fracture (Fig. 7(c)). The depth of the dimples was smaller than that of the W₀ alloy due to the solid-solution strengthening of the FCC phase. The particles at the bottom of the dimple were also Cr₂O₃. With increase in the volume fraction of the μ phase, the fracture mode of the alloys changed from ductile to brittle fracture. As shown in Fig. 7(e) (g) and (i), the tensile fracture surface of the W_{0.5}, W_{0.7}, and W_{1.0} alloys showed brittle fracture morphology

with cleavage fracture in the μ phase or μ + BCC phase. Although the propagation of several internal micro-cracks formed in the μ phase or BCC phase were arrested by the ductile FCC phase in the W_{0.5}, W_{0.7}, and W_{1.0} alloys (Fig. 7(f) (h) and (j)), the continuous μ phase contributes to the expansion of the crack, which led to a decrease in the ductility.

4. Conclusion

In this study, the CoCrFeNiW_x ($x = 0, 0.2, 0.5, 0.7$, and 1.0) HEAs with different W contents were prepared by powder plasma arc additive manufacturing (PPA-AM). The microstructure and mechanical properties of the CoCrFeNiW_x alloys were studied systematically, and the following conclusions were drawn:

- (1) The crack-free wall samples made of CoCrFeNiW_x HEAs were successfully prepared by PPA-AM using Co, Cr, Fe, Ni and W mixed powder blends. The microstructure of the alloys changed from a single FCC phase ($x = 0$) to FCC + μ phase ($x = 0.2$ and 0.5), and further to FCC + μ + BCC phase ($x = 0.7$ and 1.0), with the increase in the W content. The addition of W caused the distortion of FCC, which resulted in an increase in the lattice parameters.
- (2) The FCC phase of the samples has a strong {001} texture, while the μ phase and BCC phase show random growth. The addition of

- W significantly reduces the grain size of the FCC phase, which is mainly due to the nucleation provided by the μ and W phase for the growth of the FCC phase and the hindrance offered by large-size precipitates to the growth of the FCC phase.
- (3) The addition of W significantly increased the hardness (from 136 HV to 413 HV) and yield strength (from 187 MPa to 567 MPa) of CoCrFeNiW_x HEAs, while the ductility decreased (from 50.8–0.3%). The strengthening mechanisms included solid solution strengthening and second phase strengthening by W addition. In tensile tests, the W_{0.5}, W_{0.7}, and W_{1.0} alloys showed brittle fracture with elongations of 2.6%, 0.6%, and 0.3%, respectively, which was due to the contributing effect of the brittle μ phase to the expansion of the crack.

CRediT authorship contribution statement

Qingkai Shen: Methodology, Formal analysis, Writing – original draft. **Jiaxiang Xue:** Supervision, Project administration, Funding acquisition. **Xiaoyan Yu:** Investigation. **Zehong Zheng:** Writing – review & editing. **Ning Ou:** Validation, Data curation.

Data Availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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